

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

- A coin is tossed 10 times and the outcomes are:
H H T H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
- Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
- A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
- A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
- Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 - Find the predicted output.
 - Calculate the error.
 - If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I Kumaran.M(192525022) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

solutions:

1. Coin Toss – Maximum Likelihood Estimation (MLE)

Given outcomes:

H H T H H H T H H T

Total tosses, $n=10$

Number of Heads = 7

The MLE for the probability of Heads is:

$p = \text{Number of Heads} / \text{Total Tosses}$

$p = 7/10 = 0.7$

Answer:

$p = 0.7$

2. Students Passing an Exam

Given:

Total students = 50

Passed = 42

MLE:

$p = 42 / 50 = 0.84$

Answer:

Probability of passing = 0.84 (84%)

3. Defective Light Bulbs

Given:

Total bulbs = 100

Defective bulbs = 8

MLE:

$$p = 8 / 100 = 0.08$$

Answer:

Probability of a bulb being defective = 0.08 (8%)

4. Biased Die

Given:

Total rolls = 60

Number of times 6 appears = 18

MLE:

$$p = 18 / 60 = 0.3$$

Answer:

Probability of rolling a 6 = 0.3 (30%)

5. Prediction, Error and Weight Update

Given:

Input, $x=4$

Weight, $w=2$

Actual Output = 10

Prediction formula:

$$y = w * x$$

(i) Predicted Output

$$y = 2 \times 4 = 8$$

Answer:

Predicted Output = 8

(ii) Calculate the Error

$$\begin{aligned}\text{Error} &= \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2\end{aligned}$$

Answer:

$$\text{Error} = 2$$

(iii) Updated Weight

Given:

$$\text{Gradient} = 4$$

$$\text{Learning Rate} = 0.01$$

Gradient descent update rule:

$$\begin{aligned}w(\text{new}) &= w - (\text{Learning Rate} \times \text{Gradient}) \\ &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96\end{aligned}$$

Answer:

$$\text{Updated Weight} = 1.96$$

